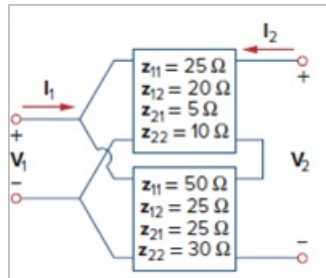


Problem

For the parallel-series connection of the two two-ports in Fig. 19.117, find the g parameters.

Figure 19.117



Step-by-step solution

Step 1 of 5

Draw the circuit diagram as shown below.

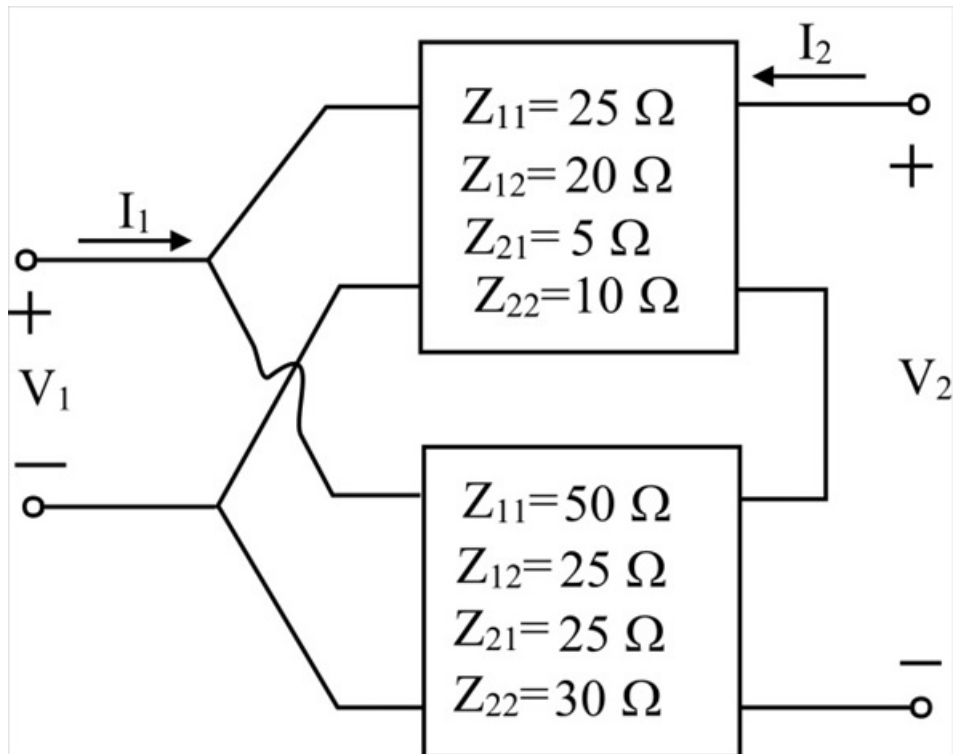


Figure 1

Step 2 of 5

Consider the following data:

$$\begin{aligned}[Z_a] &= \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \\ &= \begin{bmatrix} 25 & 20 \\ 5 & 10 \end{bmatrix}\end{aligned}$$

$$\begin{aligned}[Z_b] &= \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \\ &= \begin{bmatrix} 50 & 25 \\ 25 & 30 \end{bmatrix}\end{aligned}$$

Step 3 of 5

Calculate Δ_{Z_a} .

$$\begin{aligned}\Delta_{Z_a} &= (25)(10) - (5)(20) \\ &= 250 - 100 \\ &= 150\end{aligned}$$

Calculate Δ_{Z_b} .

$$\begin{aligned}\Delta_{Z_b} &= (50)(30) - (25)(25) \\ &= 1500 - 625 \\ &= 875\end{aligned}$$

Step 4 of 5

Calculate g -parameters from Z -parameters.

$$[g] = \begin{bmatrix} \frac{1}{Z_{11}} & -\frac{Z_{12}}{Z_{11}} \\ \frac{Z_{21}}{Z_{11}} & \frac{\Delta Z}{Z_{11}} \end{bmatrix}$$

Calculate $[g_a]$.

$$\begin{aligned}[g_a] &= \begin{bmatrix} \frac{1}{25} & -\frac{20}{25} \\ \frac{5}{25} & \frac{150}{25} \end{bmatrix} \\ &= \begin{bmatrix} 0.04 & -0.8 \\ 0.2 & 6 \end{bmatrix}\end{aligned}$$

Calculate $[g_b]$.

$$\begin{aligned}[g_b] &= \begin{bmatrix} \frac{1}{50} & -\frac{25}{50} \\ \frac{25}{50} & \frac{875}{50} \end{bmatrix} \\ &= \begin{bmatrix} 0.02 & -0.5 \\ 0.5 & 17.5 \end{bmatrix}\end{aligned}$$

Step 5 of 5

The two ports are in series.

$$\begin{aligned}[g] &= [g_a] + [g_b] \\ &= \begin{bmatrix} 0.04 & -0.8 \\ 0.2 & 6 \end{bmatrix} + \begin{bmatrix} 0.02 & -0.5 \\ 0.5 & 17.5 \end{bmatrix} \\ &= \begin{bmatrix} 0.06 \text{ S} & -1.3 \\ 0.7 & 23.5 \Omega \end{bmatrix}\end{aligned}$$

Therefore, the g -parameters are $\begin{bmatrix} 0.06 \text{ S} & -1.3 \\ 0.7 & 23.5 \Omega \end{bmatrix}$.